EXP 4

Data Scaling Using Min-Max Normalization, Standardization & Robust Scaling

AIM:

To clean the StudentsPerformance dataset by detecting and filling missing values, removing duplicate rows and making the column names consistent, and to apply Min-Max scaling, Standardization and Robust scaling to the score columns.

ALGORITHM:

1. Import pandas, numpy, matplotlib, seaborn and the MinMaxScaler, StandardScaler and RobustScaler classes from scikit-learn.
2. Load StudentsPerformance.csv into a DataFrame and display the first five rows and the shape.
3. Find the missing values in each column using isnull().sum().
4. Make the column names consistent by removing extra spaces, converting them to lower case and replacing spaces with underscores.
5. Fill the missing values of parental_level_of_education with the mode of the column and verify the result.
6. Remove the duplicate rows using drop_duplicates() and check the new shape.
7. Select the numerical columns (math_score, reading_score and writing_score) and keep a copy of the data.
8. Plot histograms of the scores before normalization.
9. Apply Min-Max scaling to the numerical columns and plot the histograms.
10. Apply Standardization (StandardScaler) to the numerical columns and plot the histograms.
11. Apply Robust scaling (RobustScaler) to the numerical columns and plot the histograms.
12. Compare the distributions before and after scaling.

CODE:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import MinMaxScaler, StandardScaler,RobustScaler
```

```
df=pd.read_csv('StudentsPerformance.csv')
```

```
df.head()
```

```
df.shape
```

```
df.isnull().sum()
```

```
df.columns = df.columns.str.strip().str.lower().str.replace(" ", "_")
```

```
df.head()
```

```
print(df.isnull().sum())
```

```
df["parental_level_of_education"] =
df["parental_level_of_education"].fillna(df["parental_level_of_education"].mode()[0])
```

```
df.isnull().sum()
```

```
df=df.drop_duplicates()
```

```
df.shape
```

```
numerical_cols = ["math_score", "reading_score", "writing_score"]

print(numerical_cols)
```

```
df_before=df.copy()
```

```
plt.figure(figsize=(15, 4))

for i, col in enumerate(numerical_cols):
    plt.subplot(1, 3, i+1)

    sns.histplot(
        df[col],
        color='red',
        kde=True
    )

    plt.title(f"Before Normalization - {col}")
    plt.xlabel(col)
    plt.ylabel("Frequency")

plt.tight_layout()
plt.show()
```

```
minmax_scaler = MinMaxScaler()

df_minmax = df.copy()

df_minmax[numerical_cols] = minmax_scaler.fit_transform(
    df[numerical_cols]
)
```

```
plt.figure(figsize=(15, 4))

for i, col in enumerate(numerical_cols, 1):
    plt.subplot(1, 3, i)

    sns.histplot(
        df_minmax[col],
        kde=True,
        color='green'
    )
```

```
    plt.title(f"Min-Max Normalized - {col}")
    plt.xlabel(col)
    plt.ylabel("Frequency")

plt.tight_layout()
plt.show()
```

```
standard_scaler = StandardScaler()

df_standard = df.copy()

df_standard[numerical_cols] = standard_scaler.fit_transform(
    df[numerical_cols]
)
```

```
plt.figure(figsize=(15, 4))

for i, col in enumerate(numerical_cols, 1):
    plt.subplot(1, 3, i)

    sns.histplot(
        df_standard[col],
        kde=True
    )

    plt.title(f"Standardized - {col}")
    plt.xlabel(col)
    plt.ylabel("Frequency")

plt.tight_layout()
plt.show()
```

```
robust_scaler = RobustScaler()

df_robust = df.copy()

df_robust[numerical_cols] = robust_scaler.fit_transform(
    df[numerical_cols]
)
```

```
plt.figure(figsize=(15, 4))

for i, col in enumerate(numerical_cols, 1):
    plt.subplot(1, 3, i)

    sns.histplot(
        df_robust[col],
        kde=True,
        color='orange'
    )

    plt.title(f"Robust Scaled - {col}")
    plt.xlabel(col)
    plt.ylabel("Frequency")

plt.tight_layout()
plt.show()
```

OUTPUT:

Output of: df.head()

```
   gender race/ethnicity parental level of education          lunch  \
0  female        group B           bachelor's degree      standard
1  female        group C                some college      standard
2  female        group B             master's degree      standard
3    male        group A          associate's degree  free/reduced
4    male        group C                some college      standard

  test preparation course   math score  reading score  writing score
0                    none           72             72             74
1               completed           69             90             88
2                    none           90             95             93
3                    none           47             57             44
4                    none           76             78             75
```

Output of: df.shape

```
(1005, 8)
```

Output of: df.isnull().sum()

```
gender                       0
race/ethnicity               0
parental level of education    7
lunch                        0
test preparation course      0
math score                   0
reading score                0
writing score                0
dtype: int64
```

Output of: df.columns = df.columns.str.strip().str.lower().str.replace(" ", "_")

```
   gender race/ethnicity parental_level_of_education          lunch  \
0  female        group B           bachelor's degree      standard
1  female        group C                some college      standard
2  female        group B             master's degree      standard
3    male        group A          associate's degree  free/reduced
4    male        group C                some college      standard

  test_preparation_course   math_score  reading_score  writing_score
0                    none           72             72             74
1               completed           69             90             88
2                    none           90             95             93
3                    none           47             57             44
4                    none           76             78             75
```

Output of: print(df.isnull().sum())

```
gender                       0
race/ethnicity               0
parental_level_of_education    7
lunch                        0
test_preparation_course      0
math_score                   0
```

```
reading_score                0
writing_score                0
dtype: int64
```

Output of: df.isnull().sum()

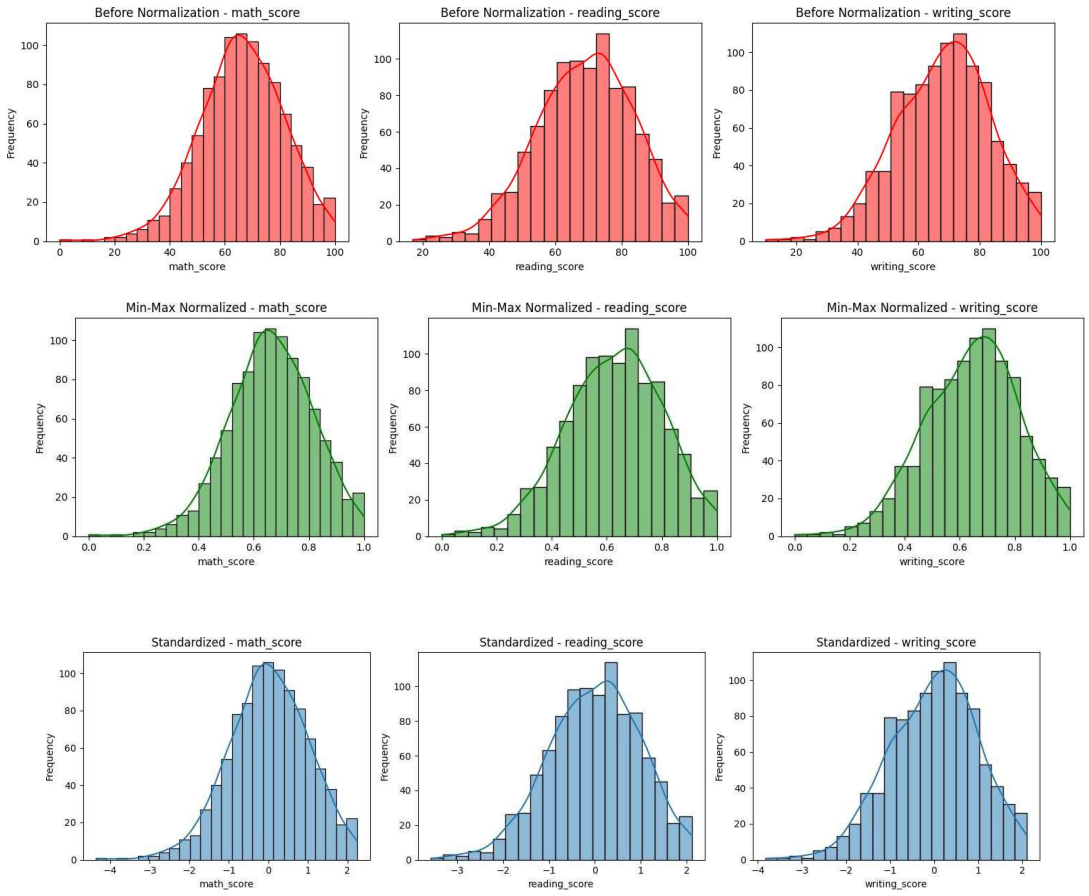
```
gender                       0
race/ethnicity               0
parental_level_of_education  0
lunch                        0
test_preparation_course      0
math_score                   0
reading_score                0
writing_score                0
dtype: int64
```

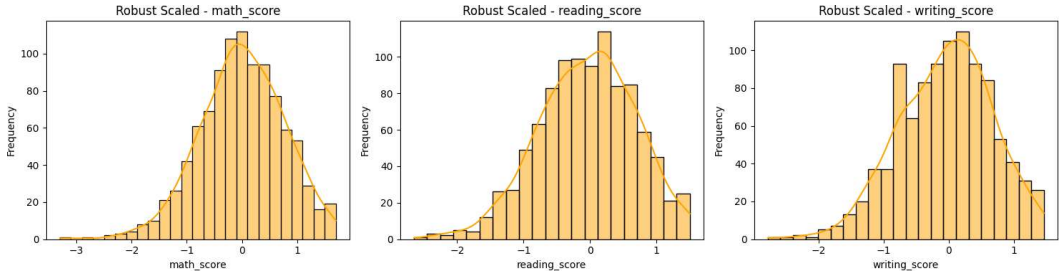
Output of: df.shape

```
(1000, 8)
```

Output of: numerical_cols = ["math_score", "reading_score", "writing_score"]

```
['math_score', 'reading_score', 'writing_score']
```





RESULT:

Thus, the Students Performance dataset was cleaned. The 7 missing values in parental level of education were filled with the mode, 5 duplicate rows were removed (1005 rows reduced to 1000), and Min-Max scaling, Standardization and Robust scaling were applied and compared using histograms.